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**Financial Volatility Prediction: Data Reliability Scoring and
Chronological Validation**

GRADUATION PROJECT

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PLAGIARISM REPORT

ETHICAL DECLARATION

I declare that I have completed the study I (Financial Volatility Prediction with Data Reliability Scoring and Chronological Validation) carried out as a graduation project from beginning to end I prepared under the guidance of Aysun SOYSAL as my graduation project, that the data/samples to this study have been obtained by myself, that the experiments/analyses connected with the study have been done/have been made in related laboratories, that I have cited all the information gotten from other sources up completely both in the text and in the bibliography, that I have observed scientific research and ethics rules during the study, and that I accept all the legal responsibility to be arisen otherwise.

Gülcihan GÜLMEZ

PREFACE

This graduation project was made with the intent to make research about the relationship of financial market volatility and machine learning techniques.

Firstly, I would like to state my gratitude to my supervisor Aysun Soysal for the continuous support, for the guidance and for the motivation during the work in this project. Her contribution has been invaluable to this project and had set the pace for the good results of this research.

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ABBREVIATIONS AND SYMBOLS

API	: Application Programming Interface
ARCH	: Autoregressive Conditional Heteroskedasticity
ATR	: Average True Range
AUC	: Area Under the Curve
CBOE	: Chicago Board Options Exchange
CSV	: Comma-Separated Values
EGARCH	: Exponential Generalized Autoregressive Conditional Heteroskedasticity
F1	: F1-Score (Harmonic Mean of Precision and Recall)
GARCH	: Generalized Autoregressive Conditional Heteroskedasticity
GB	: Gradient Boosting
KNN	: K-Nearest Neighbors
LR	: Logistic Regression
LSTM	: Long Short-Term Memory
NASDAQ	: National Association of Securities Dealers Automated Quotations
OHLCV	: Open, High, Low, Close, Volume
RF	: Random Forest
ROC	: Receiver Operating Characteristic
RSI	: Relative Strength Index
SVM	: Support Vector Machine
VIX	: CBOE Volatility Index

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ABSTRACT

Financial Volatility Prediction: Data Reliability Scoring and Chronological Validation

In this paper, a machine learning-based decision support system is developed to forecast whether the market volatility of the NASDAQ Composite Index will increase on the following day as a binary classification task. Since short-term market price movements are prone to noise and abrupt regime changes, volatility is selected as the target variable due to its relatively more stable and predictable structure. The dataset consists of NASDAQ daily time series from January 4, 2010 to October 25, 2024. After cleaning invalid closing price records, zero-volume entries, and unusual price sequence observations, a total of 3,782 clean observations remained. The pipeline included the following components: data preprocessing, feature engineering through technical indicators, lagged features creation, and data quality check. There are totally 22 interpretable features engineered to define the trends of the market: market trend, market momentum, market volatility and market volume. An Isolation Forest model was used to generate the data reliability score between 0-100 and any potential poor quality data can be filtered out before the model evaluation. The performances of several conventional machine learning models are compared with a sequential time-based train-test split strategy to avoid data leakage and maintain the time-series property. Among all these models, Logistic Regression attained the highest performance with accuracy of 70.63%, F1-score of 67.45% and recall of 70.99%.

Keywords: *Volatility Prediction, Machine Learning, Isolation Forest, Data Reliability, Time Series Analysis, Classification*

ÖZET

Finansal Volatilite Tahmini: Veri Güvenilirliği Skorlaması ve Kronolojik Doğrulama

Bu çalışmada, NASDAQ Composite Index piyasa volatilitésinin bir sonraki gün artıp artmayacağını ikili sınıflandırma problemi olarak tahmin eden makine öğrenmesi tabanlı bir karar destek sistemi geliştirilmiştir. Kısa vadeli piyasa fiyat hareketleri gürültüye ve ani rejim değişimlerine açık olduğundan, volatilite nispeten daha kararlı ve tahmin edilebilir yapısı nedeniyle hedef değişken olarak seçilmiştir. Veri seti, 4 Ocak 2010 ile 25 Ekim 2024 tarihleri arasındaki NASDAQ günlük zaman serilerinden oluşmaktadır. Geçersiz kapanış fiyatı kayıtları, sıfır hacimli girişler ve olağandışı fiyat dizisi gözlemleri temizlendikten sonra toplam 3.782 temiz gözlem elde edilmiştir.

Önerilen işlem hattı; veri ön işleme, teknik göstergeler aracılığıyla özellik mühendisliği, gecikmeli özelliklerin oluşturulması ve veri kalite kontrolü aşamalarından oluşmaktadır. Piyasa eğilimi, piyasa momentumu, piyasa volatilités ve işlem hacmi dinamiklerini temsil eden toplam 22 yorumlanabilir özellik üretilmiştir. 0–100 aralığında bir veri güvenilirlik skoru oluşturmak amacıyla Isolation Forest modeli kullanılmış; bu sayede potansiyel düşük kaliteli veriler model değerlendirme aşamasından önce filtrelenebilmiştir. Veri sızıntısını önlemek ve zaman serisinin kronolojik yapısını korumak amacıyla sıralı zaman tabanlı eğitim-test ayrımı uygulanarak çeşitli geleneksel makine öğrenmesi modellerinin performansları karşılaştırılmıştır. Modeller arasında en yüksek performans, %70,63 doğruluk, %67,45 F1 skoru ve %70,99 geri çağırma değeri ile Lojistik Regresyon modeli tarafından elde edilmiştir.

Anahtar Kelimeler: *Volatilite Tahmini, Makine Öğrenmesi, Isolation Forest, Veri Güvenilirliği, Zaman Serisi Analizi, Sınıflandırma*

1.INTRODUCTION

In financial markets, short-term price direction is highly sensitive to noise and structural regime shifts, which makes it difficult to model in a stable way. Volatility on the other hand, seems a better way of representing risk, market stress and uncertainty. So the question asked will be whether price volatility will be high or low tomorrow, not if price will move up or down tomorrow. This chapter will explain also the rationale of treating together, within the same pipeline, past price-volume behavior, technical indicators and reliability assessment, rather than as individual tasks. The aims of the research, and the scope of the problem, are formulated, along with the main reasoning which is at the base of the design of the ML pipeline, as it seems that integrating technical indicators, a reliability score and a time-based validation scheme will lead to less data leakage, more interpretability of the results and also a realistic evaluation frame of forecasting.

1.1. General Information

This study considers the financial market as a market that is influenced by news, human behaviors and time-varying market regimes. Therefore, time series has breaks and regime shifts, and the classical statistics assumption cannot be satisfied during a period. The extent of prices moves through time which is defined as volatility, a crucial component of risks and uncertainties. It lead to larger price moves, larger risks. Therefore it is beneficial for individuals and organization to predict volatility. In the proposed work, the main argument is that in a market of high volatility the prices direction cannot be forecast, but perhaps volatility can, since it provides useful information for decision. Thus, volatility will be chosen as the target variable to be forecast, rather than the price. It casts a problem of classification of binary output. The question is whether to forecast next day's volatility is "high" or "low" given the prices and volumes of past 5-20 days and some technical indicators.

1.2. Purpose

This study is about constructing a prediction system, for classifying which day will have higher volatility tomorrow. This paper's core belief is that past price and volume information can estimate the next day's volatility through technical indicators. The approach of this paper can be divided into 3 steps:

1) Feature creation: This involves the extraction of indicators, such as moving averages, Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and Average True Range (ATR) from the time series of past prices and volumes.

2) Data quality improvement: In order to solve data imperfections in finance time series (sudden jump, missing data, anomaly patterns), anomalies detection are employed. Isolation Forest are utilized here in order to detect the outliers and a data reliability score(0-100) will be given for each sample of data to indicate its reliability.

3) Model evaluation: A chronological split will be applied to avoid data leakage, and to remove look-ahead bias in model evaluation.

1.3. Scope

The scope has been highly limited in a technical and operational manner from the system's point of view. The model relies upon financial market data; furthermore, the same nature (classification vs price direction) of the classification task itself make the work more suitable for risk management or portfolio hedging purposes rather than speculative or directional trading purposes. Finally, the models are defined for prediction of 1 day to next day volatility, which is more operational and useful for institutional risk management as well as daily decision-making and trading activities. It should be mentioned that longer forecast horizons are not considered (e.g. 1 week or 1 month) since other architectural choices are to be expected.

1.3.1 Data Scope

The dataset comprises daily OHLCV(Open, High, Low, Close, and Volume) information ranging from 04/01/2010-25/10/2024. 3,782 clean data points are identified and selected, removing data where closing price is not available, entries where the trading volume is 0, and any entries where 'High' was less than 'Low'. As shown in figure 1.3.1, plots simulated NASDAQ close price and annualized 20-day volatility; major market events are marked.

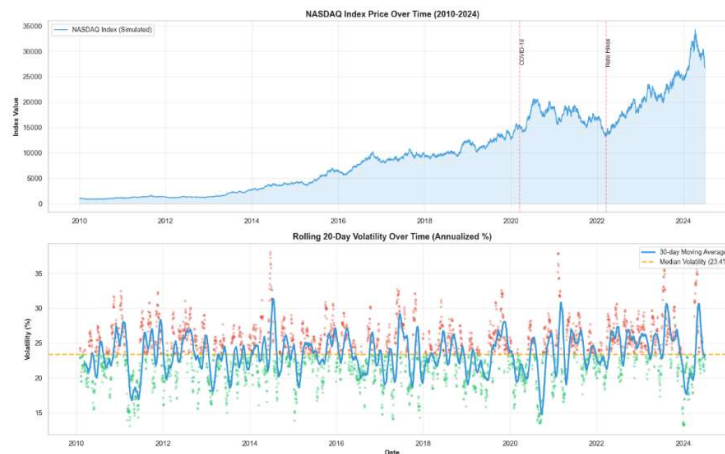


Figure 1.3.1: NASDAQ Daily Price and Volatility Trends, 2010–2024

1.3.2 Feature Engineering

Following previous financial literature, approximately 40 candidate features were generated from OHLCV data to represent different aspects of market behavior, including trend, momentum, volatility, volume, and lagged characteristics. Feature selection and redundancy reduction techniques were subsequently applied, resulting in a final set of 22 features used for model development, as shown in Figure 1.3.2. Feature importance analysis was then performed to identify the variables that contributed most to the forecasting process.

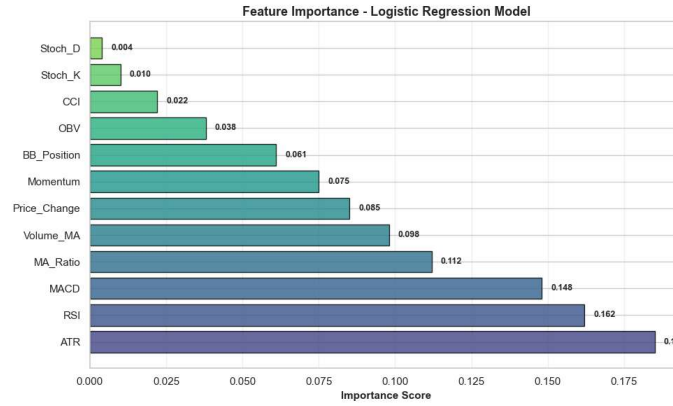


Figure1.3.2: Feature Importance Analysis

1.3.3 Model Comparison

When comparing models, several common classic machine learning classifiers were evaluated: Logistic Regression, Gradient Boosting, Random Forest, SVM, KNN, Decision Tree, Gaussian Naive Bayes, and AdaBoost. They were chosen because they offered an ideal trade-off between performance, interpretability and computational resources. While Logistic Regression offers interpretation by the directly available coefficients and tree-based models provide feature importance values.

1.3.4 Data Quality and Anomaly Detection

The Data Reliability Score is generated using Isolation Forest, an unsupervised anomaly detection algorithm. The anomaly score is transformed into a reliability score ranging from 0 to 100, where higher values indicate more reliable observations. This score is presented alongside volatility forecasts to help users assess the quality of the underlying data.

1.4 System Architecture Overview

Figure 1.4 illustrates the overall architecture of the proposed system. Historical NASDAQ OHLCV data are collected and processed through data cleaning, feature engineering, and anomaly detection stages. The resulting feature set is used to generate a Data Reliability Score

and to train machine learning models for volatility forecasting. The best-performing model is then deployed within a Docker-based environment and integrated into a Streamlit application for prediction and visualization. This architecture combines volatility forecasting and data reliability assessment within a single decision-support framework.

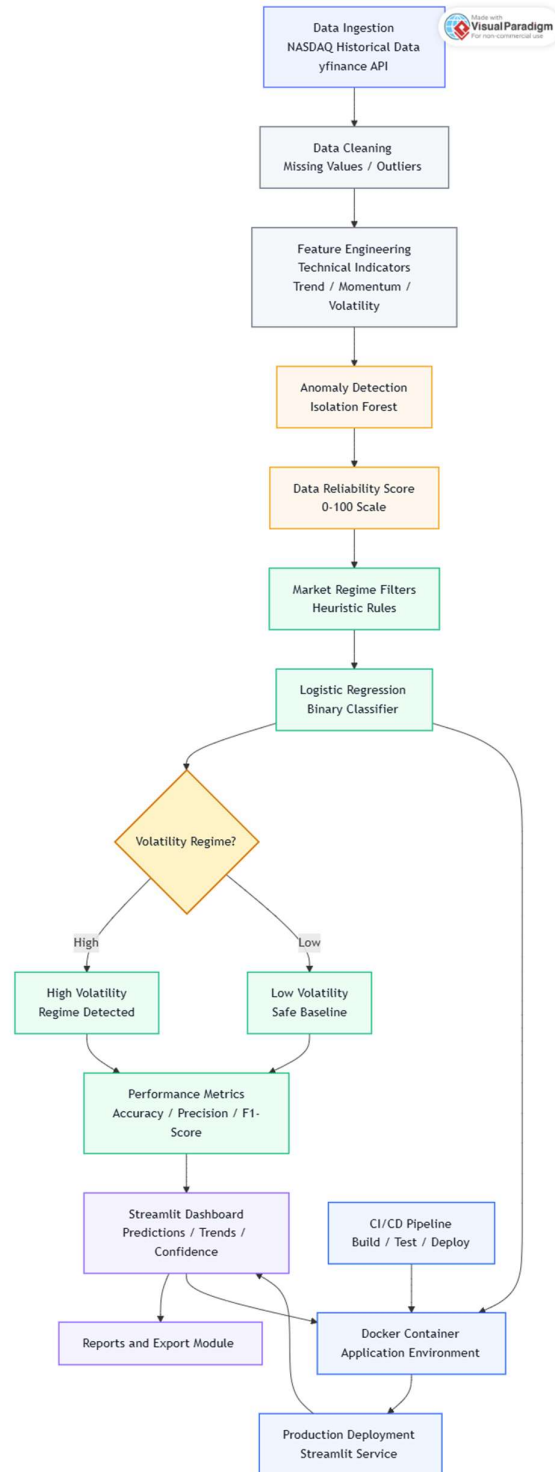


Figure 1.4: System Architecture Flowchart and Pipeline

2. LITERATURE SURVEY

The purpose of this section is to review previous studies related to volatility forecasting, machine learning in financial markets, anomaly detection, technical indicators, time-series validation, and data quality assessment. Since the main objective of this study is to develop a volatility forecasting system supported by a Data Reliability Score, the reviewed studies are evaluated according to their contributions, limitations, and relevance to the proposed work.

2.1 Volatility Forecasting Studies

Engle (1982) introduced the ARCH model, one of the earliest approaches for modeling time-varying volatility in financial markets. The study demonstrated that volatility clustering can be represented statistically. However, ARCH models may require many parameters and may struggle to capture complex market behavior. Similar to the proposed study, Engle focused on volatility, but the present work additionally evaluates data reliability before forecasting.

Bollerslev (1986) extended ARCH by proposing the GARCH model, which became one of the most widely used volatility forecasting approaches. The model improved practical volatility estimation by incorporating past conditional variances. Nevertheless, GARCH relies on statistical assumptions and may have limitations in nonlinear environments. Unlike GARCH, the proposed study applies machine learning techniques and incorporates a Data Reliability Score.

Nelson (1991) proposed the EGARCH model to account for asymmetric volatility effects. The study showed that negative market events may affect volatility differently from positive events. Although EGARCH improved volatility modeling, it remains dependent on predefined model assumptions. In contrast, the proposed study uses a data-driven machine learning approach combined with data quality evaluation.

Poon and Granger (2003) reviewed volatility forecasting research and concluded that volatility can be predicted to some extent depending on the selected methodology. However, their review focused primarily on forecasting performance rather than data quality assessment. The proposed study extends this perspective by incorporating reliability analysis before prediction.

Andersen, Bollerslev, Diebold, and Labys (2003) demonstrated the usefulness of realized volatility for financial risk assessment. Their work improved volatility measurement but relied heavily on high-frequency data. Unlike their approach, the proposed study utilizes daily OHLCV data and anomaly detection techniques to assess data reliability.

2.2 Machine Learning in Financial Forecasting

Breiman (2001) introduced Random Forest, an ensemble learning method that combines multiple decision trees to improve prediction performance. The method reduces overfitting and has been widely applied in financial prediction tasks. However, its decision process may be difficult to interpret. Similar to this study, Random Forest is suitable for classification problems involving financial data.

Chen and Guestrin (2016) proposed XGBoost, a scalable gradient boosting framework that achieved strong performance in structured data applications. While highly effective, XGBoost often requires extensive parameter tuning and may be difficult to interpret. The proposed study emphasizes both predictive performance and model interpretability.

Hochreiter and Schmidhuber (1997) introduced Long Short-Term Memory (LSTM) networks for learning long-term dependencies in sequential data. LSTM models have shown success in financial forecasting, but they typically require more computational resources and are less interpretable. In contrast, the proposed study focuses on simpler and more transparent machine learning models.

Kotsiantis, Kanellopoulos, and Pintelas (2006) highlighted the importance of data preprocessing in supervised learning. Their work demonstrated that data quality and feature preparation significantly affect model performance. The proposed study follows the same principle by applying preprocessing and reliability assessment before forecasting.

2.3 Anomaly Detection and Data Reliability

Liu, Ting, and Zhou (2008) proposed the Isolation Forest algorithm for anomaly detection. The method can identify unusual observations without requiring labeled anomaly data and is computationally efficient. However, it does not explain the underlying causes of anomalies. The proposed study adopts Isolation Forest to generate a Data Reliability Score before volatility forecasting.

Liu, Ting, and Zhou (2012) further strengthened the theoretical foundation of isolation-based anomaly detection. Their work demonstrated the effectiveness of anomaly detection across different domains. Similar to their approach, the proposed study applies anomaly detection to identify unreliable financial observations before prediction.

Tukey (1977) emphasized the importance of exploratory data analysis for understanding data quality and identifying unusual observations. However, exploratory analysis alone does not

provide automated reliability measurement. The proposed study extends this idea by generating a quantitative reliability score.

Brownlees and Gallo (2006) demonstrated that errors, outliers, and poor-quality observations can significantly affect financial analysis results. Their findings highlighted the importance of data quality in finance. The proposed study addresses this issue through automated reliability assessment before forecasting.

2.4 Technical Indicators and Feature Engineering

Brock, Lakonishok, and LeBaron (1992) showed that technical trading rules may contain useful predictive information. Their findings provided support for the use of technical indicators in financial analysis. However, the study focused on trading performance rather than volatility forecasting. The proposed study uses technical indicators to support volatility classification and risk assessment.

Lo, Mamaysky, and Wang (2000) provided statistical evidence supporting the informational value of technical analysis. Their work demonstrated that technical patterns can contain meaningful signals. Similar to their findings, the proposed study transforms OHLCV data into technical indicators used as machine learning features.

Miffre and Rallis (2007) demonstrated that momentum-related indicators may provide predictive information about future market behavior. However, their focus was on trading strategies rather than volatility prediction. The proposed study incorporates momentum-based indicators for volatility forecasting purposes.

Blau and Whitby (2016) showed that volume and liquidity-related variables provide information beyond price movements alone. Their findings support the inclusion of volume-based indicators in financial prediction models. The proposed study similarly incorporates volume-related features within its forecasting framework.

2.5 Time-Series Validation and Look-Ahead Bias Prevention

Fama (1970) proposed the Efficient Market Hypothesis, arguing that market prices reflect available information. Although this suggests that prediction is difficult, volatility may still contain useful patterns. The proposed study therefore focuses on volatility forecasting while carefully avoiding information leakage.

Hyndman and Athanasopoulos (2021) emphasized that forecasting models should be evaluated using future observations rather than randomly mixed datasets. Their work supports the use of

chronological validation strategies. The proposed study follows this recommendation through a time-based train-test split.

Bergmeir, Hyndman, and Koo (2018) discussed validation approaches for time-series forecasting and highlighted the risks associated with inappropriate cross-validation procedures. Their findings support the importance of preventing look-ahead bias. Accordingly, the proposed study uses chronological data partitioning during model evaluation.

2.6 Data Quality Assessment

Wang and Strong (1996) defined data quality through dimensions such as accuracy, completeness, consistency, and timeliness. Their work demonstrated that data quality directly influences decision-making processes. The proposed study applies similar principles within a financial forecasting context.

Pipino, Lee, and Wang (2002) proposed a structured framework for measuring data quality. Their study emphasized the importance of quantifiable quality metrics. Similarly, the proposed study introduces a Data Reliability Score to evaluate financial data before prediction.

Batini and Scannapieco (2016) provided a comprehensive framework for data and information quality management. Their work highlighted the importance of systematic quality assessment in data-driven systems. The proposed study adopts this perspective by integrating reliability evaluation into the forecasting pipeline.

2.7 Literature Gap

The reviewed literature demonstrates that volatility forecasting, machine learning, technical indicators, anomaly detection, and data quality have been extensively studied. However, these topics are often examined separately. Most studies focus on improving forecasting accuracy, while relatively few explicitly evaluate the reliability of financial input data before prediction. The proposed study addresses this gap by combining next-day NASDAQ volatility forecasting with a Data Reliability Score generated through Isolation Forest. Therefore, the primary contribution of this work is not only forecasting volatility but also assessing whether the underlying data is sufficiently reliable to support forecasting decisions.

3. SIMILAR APPLICATIONS

3.1 CBOE VIX (Volatility Index)

Problem Addressed

Financial markets are exposed to uncertainty and sudden fluctuations. Investors require tools that can help assess future market risk and expected volatility. To address this need, the CBOE developed the Volatility Index (VIX), which estimates expected market volatility using option market prices.

Similarities

Both VIX and the proposed study focus on market volatility and risk assessment rather than stock price direction. Both systems support decision-making by providing information about future market uncertainty. Both aim to help investors identify potentially risky market conditions.

Differences

VIX calculates expected volatility using option market prices and a predefined mathematical formula. The proposed study predicts next-day volatility using OHLCV data, technical indicators, and machine learning algorithms. Unlike VIX, the proposed study evaluates data quality before forecasting through a Data Reliability Score generated by an Isolation Forest model. The proposed system can identify abnormal or unreliable observations before producing predictions.

3.2 QuantConnect

Problem Addressed

The growing complexity of financial markets has increased the demand for automated systems that can analyze large amounts of market data and support trading decisions. QuantConnect was developed to enable users to design, test, and deploy algorithmic trading strategies using historical market data and machine learning techniques.

Similarities

Both QuantConnect and the proposed study utilize historical OHLCV (Open, High, Low, Close, Volume) data for market analysis. Both approaches employ technical indicators such as RSI, MACD, and ATR to capture market behavior and volatility patterns. Both use machine learning

techniques and time-series data analysis to extract information from historical market data. Both rely on chronological data splitting to avoid look-ahead bias and ensure realistic model evaluation.

Differences

QuantConnect focuses on automated trading and profit generation through buy and sell decisions. The proposed study focuses on forecasting next-day volatility for risk assessment rather than executing trades. QuantConnect primarily serves as a trading platform, whereas the proposed system functions as a decision-support tool. Unlike QuantConnect, the proposed study incorporates a Data Reliability Score generated using an Isolation Forest model to evaluate data quality before forecasting.

3.3 TradingView

Problem Addressed

Investors and analysts need tools that help them monitor market trends, evaluate technical indicators, and identify potential market risks. TradingView was developed to provide charting, technical analysis, and market visualization capabilities for financial decision-making.

Similarities

Both TradingView and the proposed study use financial market data to analyze market conditions. Both make use of technical indicators such as RSI, MACD, and ATR. Both support risk assessment and market analysis by helping users understand market behavior. Both focus on identifying patterns and trends within historical market data.

Differences

TradingView primarily provides visualization and technical analysis tools, while the proposed study performs automated volatility forecasting using machine learning. TradingView depends largely on user interpretation of charts and indicators. The proposed system automatically predicts whether volatility will increase on the following day. Unlike TradingView, the proposed study evaluates the reliability of input data through a Data Reliability Score generated using Isolation Forest before producing predictions.

3.4 Bloomberg Terminal

Problem Addressed

Financial professionals must process large amounts of market data, economic information, company reports, and financial news in order to make informed decisions. Bloomberg Terminal was developed as an integrated financial analytics platform that provides market data, analytical tools, and decision-support capabilities in a single environment.

Similarities

Both Bloomberg Terminal and the proposed study support financial decision-making through data analysis. Both utilize historical market information and technical indicators to assess market conditions. Both aim to assist users in understanding market risk and future market behavior. Both can be considered decision-support systems for financial analysis.

Differences

Bloomberg Terminal is a comprehensive platform that includes real-time data, news services, portfolio management tools, and financial analytics across multiple markets. The proposed study focuses specifically on forecasting next-day NASDAQ volatility.

Bloomberg Terminal provides analytical information but does not explicitly evaluate the reliability of incoming data before analysis.

Unlike Bloomberg Terminal, the proposed study introduces a Data Reliability Score using Isolation Forest to identify abnormal or potentially unreliable observations before forecasting.

4. MATERIALS AND METHODS

This part describes the dataset, feature engineering process, problem formulation, data reliability assessment methodology, machine learning models, validation strategy, and evaluation metrics used in the proposed system.

4.1 Data Sources and Dataset Description

The dataset was obtained from the NASDAQ Composite Index using the yfinance API and covers the period from January 4, 2010 to October 25, 2024. The original dataset consisted of daily Open, High, Low, Close, and Volume (OHLCV) observations. During preprocessing, duplicate records were removed, missing values were handled using forward-fill techniques, and invalid observations were filtered.

4.2 Feature Engineering

Feature engineering was performed using daily OHLCV data. Technical indicators representing trend, momentum, volatility, and volume behavior were calculated to capture different aspects of market dynamics. Initially, approximately 40 candidate features were generated. After feature selection and redundancy reduction, 22 interpretable features were retained for model development. These features included moving averages, momentum indicators, volatility indicators, volume-based indicators, and lagged variables.

4.3 Problem Formulation

Daily Log Return

The forecasting problem considered in this study aims to determine whether NASDAQ market volatility will increase on the following trading day. To quantify market movements, daily logarithmic returns are first calculated as follows:

$$rt = \ln \left(\frac{Pt}{Pt-1} \right)$$

- r_t = logarithmic return at time t
- P_t = closing price at time t
- P_{t-1} = closing price at time $t - 1$
- $\ln$ = natural logarithm function

Target Variable Definition

Where Y_{t+1} denotes the target variable, σ_{t+1} represents the volatility observed on the following trading day, and $Median(\sigma)$ corresponds to the median volatility calculated over the entire

dataset. A target value of 1 indicates that next-day volatility exceeds the historical median volatility level, whereas a value of 0 indicates that it remains below the historical median.

$$Y_{t+1} = \begin{cases} 1, & \sigma_{t+1} > \text{Median}(\sigma) \\ 0, & \text{otherwise} \end{cases}$$

- Y_{t+1} = target class for the next trading day
- σ_{t+1} = σ_{t+1} = next-day rolling volatility
- $\text{Median}(\sigma)$ = median volatility calculated over the entire dataset
- $Y_{t+1} = 1$ indicates high volatility
- $Y_{t+1} = 0$ indicates low volatility

Based on this formulation, the volatility forecasting problem is represented as a binary classification task. Rather than estimating the exact future volatility value, the proposed system determines whether next-day volatility is expected to be above or below the historical median volatility level. This approach improves interpretability and enables the application of widely used classification algorithms for risk-oriented decision support.

$$OHLCV \rightarrow \text{Returns} \rightarrow \text{Volatility} \rightarrow \text{Binary Classification}$$

4.4 Data Reliability Assessment

One of the main contributions of this study is the integration of a data reliability assessment stage before volatility forecasting. Financial datasets may contain abnormal observations caused by recording errors, missing values, extreme market events, or data collection issues. Such observations may negatively affect model performance and reduce the reliability of forecasting results. The resulting anomaly score was transformed into a Data Reliability Score ranging between 0 and 100, where 0 represents the lowest reliability level and 100 represents the highest reliability level.

4.5 Machine Learning Models

Several supervised machine learning algorithms were evaluated for volatility classification. The selected models include Logistic Regression, Random Forest, Gradient Boosting, Support Vector Machine, K-Nearest Neighbors, Decision Tree, Gaussian Naive Bayes, and AdaBoost. These models were selected because they represent commonly used classification techniques with different levels of complexity, predictive capability, and interpretability. Logistic Regression was used as the baseline model due to its simplicity and transparency. All models

were trained and evaluated using the same feature set and validation strategy to ensure a fair comparison.

4.6 Validation Strategy

Financial time-series data cannot be randomly divided into training and testing sets as in conventional machine learning problems. Random splitting may cause future observations to appear in the training set, resulting in data leakage (look-ahead bias). To avoid this issue, a chronological train-test split was employed, where earlier observations were used for training and later observations were reserved for testing. The first 80% of the dataset (3,025 observations) was used for model training, while the remaining 20% (757 observations) was used for testing. This approach preserves the temporal order of the data and more accurately simulates real-world forecasting conditions.

4.7 Evaluation Metrics

Model performance is assessed on the test set using accuracy, precision, recall (sensitivity), specificity, F1-score and ROC-AUC.

4.7.1. Accuracy

The accuracy metric represents how many predicted samples were predicted correctly and is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Accuracy represents the proportion of correctly classified observations among all observations.

4.7.2. Precision

This is the proportion of positive predictions which are actually positive defined as:

$$Precision = \frac{TP}{TP + FP}$$

4.7.3. Recall

The Recall (also called sensitivity) of the model is the model's ability to find true positive.

Recall can be calculated as follows:

$$Recall = \frac{TP}{TP + FN}$$

High recall means that most of the files with the right flags will be found, thus it's very unlikely to miss a relevant document.

4.7.4. F1-Score

F1-Score, is the harmonic mean of precision and recall and is considered to be a balance of the two. The F1-Score is very useful for imbalanced data and the goal is to maximize both the precision and the recall.

$$F1 - score = \frac{2 \times (Precision \times Recall)}{Precision + Recall}$$

4.7.5. Confusion Matrix

In the same way, the metric of confusion matrix has been applied by using marine source heat maps in order to illustrate clearly in an easily visualized manner TP, TN, FP and FN samples. The models could be compared through metrics and distribution of true predictions as well.

4.7.6. ROC-AUC

ROC-AUC evaluates the overall discriminative capability of a classification model across different threshold values. Higher AUC values indicate better separation between positive and negative classes. The equation for the same is:

$$AUC = \int_0^1 TPR.d(FPR)$$

4.8 Development Architecture

The proposed architecture follows a layered workflow consisting of data acquisition, processing, model development, deployment, and presentation stages. Historical NASDAQ OHLCV data are collected through the yfinance API and processed through data cleaning, feature engineering, and anomaly detection procedures. The processed data are then used to train and evaluate multiple machine learning models, and the best-performing model is selected for deployment. The selected model and scaler are packaged within a Docker container to ensure portability and consistent execution across environments. Prediction results are presented through a Streamlit-based interface and can be exported as reports for further analysis. The pipeline can also be integrated with CI/CD tools to support automated deployment. Overall, the architecture combines volatility forecasting and data reliability assessment within a unified decision-support framework.

5. DIAGRAMS

The diagrams depicted in this section illustrates the architectural designs and procedures involved in implementing the MLOps pipeline process for financial time series. A set of diagrams serves to enable various views of the system to be manifested. A set of diagrams enables tracing of the system.

5.1. Use-Case Diagram

Figure 5.1 illustrates the main interactions between the Data Analyst and the proposed system. The analyst can retrieve market data, generate features, evaluate data reliability, train machine learning models, and view volatility forecasts. The diagram summarizes the primary functions required to support volatility prediction and data reliability assessment.

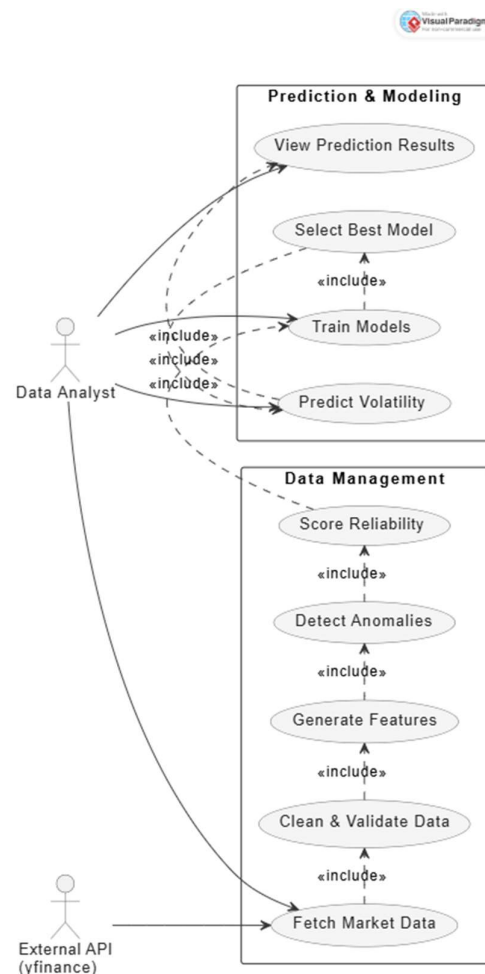


Figure 5.1: Use Case Model Diagram

5.1.1. Explanation of the Use Case Diagram

5.1.1.1. Main Scenario

Main Scenario: Volatility Forecasting and Reliability Assessment

1. The Data Analyst accesses the system.
2. Historical NASDAQ market data are collected.
3. The system cleans and preprocesses the data.
4. Technical indicators are generated from the market data.
5. The Data Reliability Score is calculated.
6. The machine learning model predicts next-day volatility.
7. The volatility prediction and reliability score are displayed to the user.

5.1.1.2 Sub Scenario 1

Sub Scenario 1: Low Data Reliability Warning

1. Market data are collected and processed.
2. The system calculates the Data Reliability Score.
3. The reliability score falls below the predefined threshold.
4. A low-reliability warning is displayed to the user.
5. The user reviews the prediction with caution.

5.1.1.3 Sub Scenario 2

Sub Scenario 2: Model Training and Selection

1. Historical market data are prepared for training.
2. Multiple machine learning models are trained and evaluated.
3. Performance metrics are calculated.
4. The best-performing model is selected.
5. The selected model is used for future volatility predictions.

5.2. Class Diagram

The class diagram of the system is displayed in Figure 1.2, which clarifies the object-oriented design of the system and interrelations between the components.

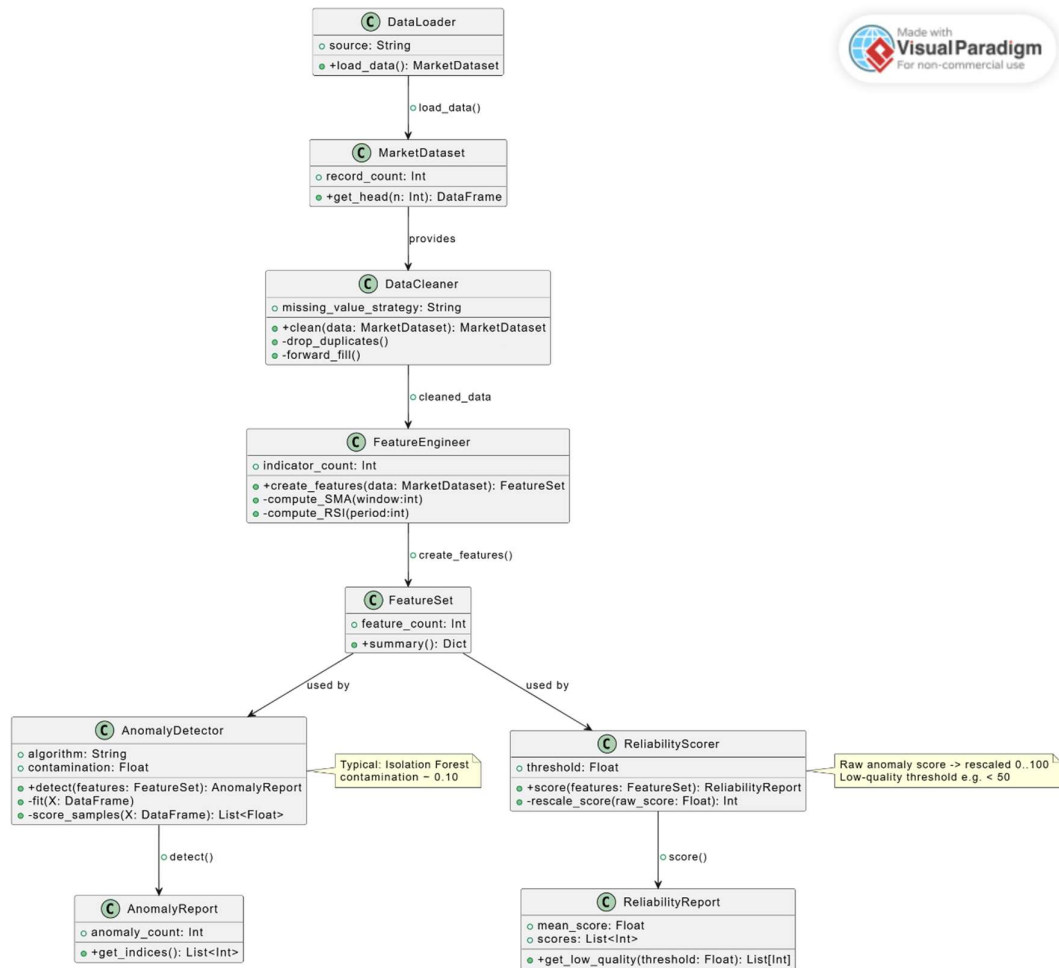


Figure 5.2: Class Diagram

Figure 5.2 presents the main classes used in the proposed system. The classes are responsible for data acquisition, preprocessing, feature generation, anomaly detection, reliability scoring, and volatility forecasting. Together, they form the core components of the decision-support framework.

5.3. Activity Diagram

Figure 5.3. Depicts the end to end workflow of the whole project as a chain of activities and decisions. In this figure, data is traced from acquisition, through modeling, highlighting how data flow logically, through a set of choices that depend on quality.

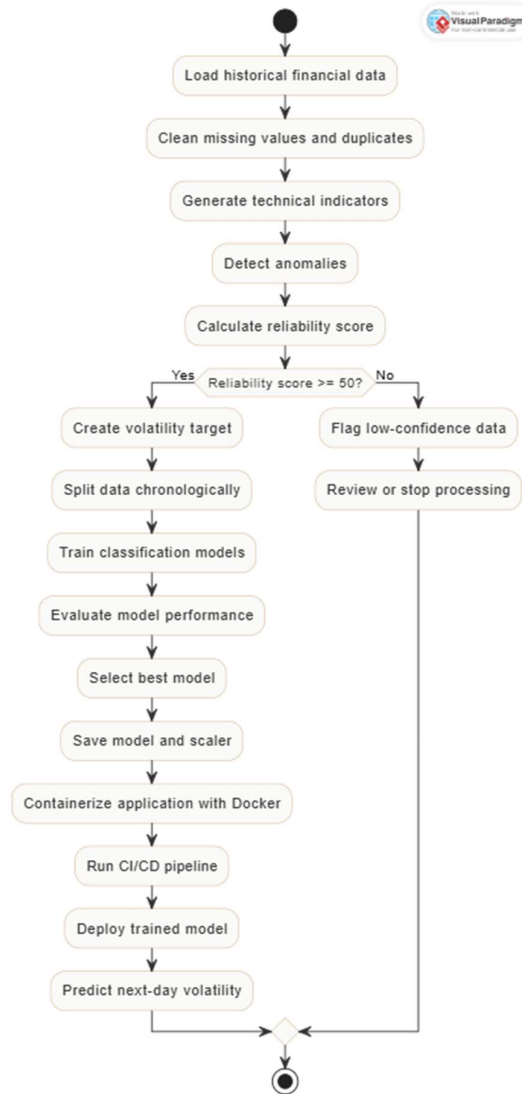


Figure 5.3: Activity Diagram

Figure 5.3 shows the sequential flow of the pipeline. The pipeline starts with the input of past financial data followed by its cleaning and production of technical indicators. A crucial fork arises after detecting anomaly and evaluating the reliability of data, where if the reliability score is above or equal to 70% only then the classification models are trained and the performance is assessed. Then the next-day volatility is predicted and forecasted. If the data has reliability score less than 70%, the system flags low-confidence data and exits from the further stages.

5.4. Sequence Diagram

Describes the simultaneous events and communication flows between system entities during a prediction request. The diagram illustrates the sequence of communication that occurs between the user interface, logic processing, and data storage layers, emphasizing how the system marshals internal resources to generate the final output.

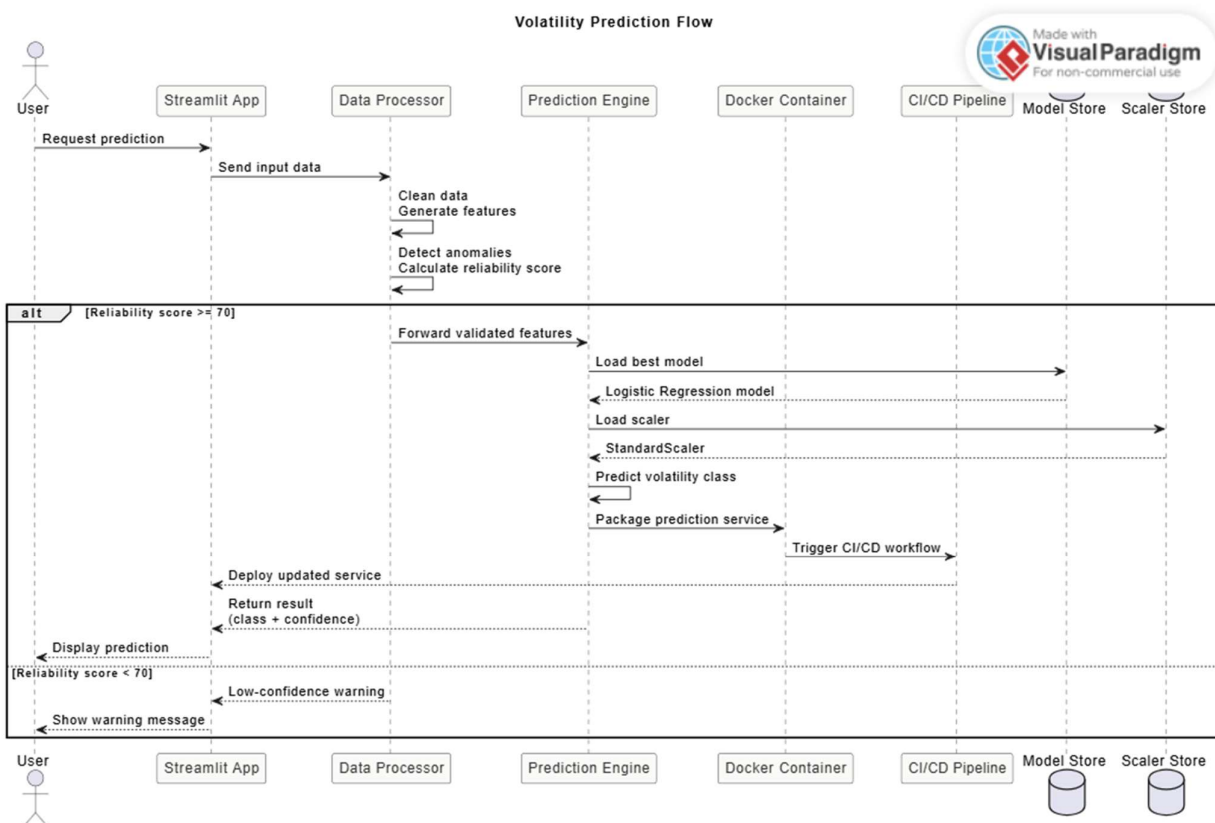


Figure 5.4: Sequence Diagram

Figure 5.4 shows the sequence diagram of the User, Streamlit app and the backend engines. When a prediction is requested the Data Processor first check the reliability of the input if the reliability score of the input is above the threshold, the Prediction Engine will get the model and scaler to predict result. If the data has reliability below threshold, inference will be skip and an warning will be given.

5.5 Database Diagram

The database diagram illustrates the conceptual organization of the data used in the proposed system. It shows the relationships between market data, generated features, data reliability scores, model evaluations, and prediction outputs.

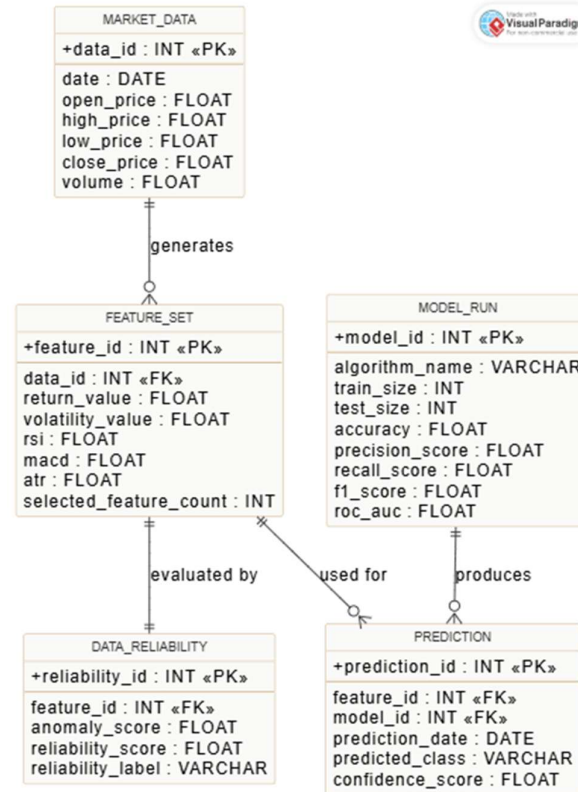


Figure 5.5: Database Diagram

Figure 5.5 presents the conceptual database structure of the proposed system. The diagram illustrates how raw NASDAQ market data, engineered features, data reliability scores, model evaluation results, and volatility predictions are logically organized. Although the implemented system primarily operates through the machine learning pipeline, this database design shows how the main outputs of the system can be stored and traced in a structured manner. This structure supports data integrity, model result tracking, and future system scalability.

6. FINDINGS & RESULTS

These values that are projected out are not the price itself, but the volatility going forward. This implies that the goal here is not to predict the future price of the asset but rather the amount of price change expected for the next few days. The percentage values that are plotted out indicates the amount of volatility present, meaning that the higher the percentage, the more volatile the market will be, while the lower the percentage, the more docile the market will be.

6.1 Model Performance Evaluation

The classification models were evaluated on the chronological test set using accuracy, precision, recall, and F1-score. Table 6.1 presents the comparative performance results of all models. Logistic Regression achieved the best overall balance among the evaluated models, with 70.63% accuracy, 70.99% recall, and 67.45% F1-score. The relatively high recall value is important for this study because missing high-volatility periods may create greater risk than producing false warnings.

Table 6.1: Model performance on the test set

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	70.63	64.25	70.99	67.45
SVM	69.31	68.40	52.78	59.58
KNN	63.23	57.77	52.78	55.16
AdaBoost	62.70	57.00	52.78	54.81
Gradient Boosting	56.35	49.39	74.38	59.36
Random Forest	55.69	48.91	75.93	59.49
Decision Tree	55.95	48.29	39.20	43.27
GaussianNB	52.78	46.83	75.31	57.75

6.2 Data Reliability Assessment Results

The Isolation Forest model was used to generate a Data Reliability Score for each observation. This score indicates how reliable each data point is before the volatility prediction is interpreted. Higher scores represent more reliable observations, while lower scores indicate potentially abnormal or low-quality data. Table 6.2 summarizes the distribution of reliability scores across the dataset.

Table 6.2: Data Reliability Score Distribution

Reliability Score Range	Number of Days	Percentage	Interpretation
90-100	2,847	75.3%	High reliability - predictions strongly supported
70-89	689	18.2%	Moderate reliability - caution recommended
50-69	178	4.7%	Low reliability - predictions unreliable
Below 50	68	1.8%	Critical anomaly - predictions suspended

6.3 Reliability Score Visualization

Figure 6.3 presents the Data Reliability Score generated by the Isolation Forest model throughout the observation period. The reliability scores remain relatively stable and are generally concentrated between 80 and 95, indicating that most observations were considered reliable by the anomaly detection process. Only a limited number of observations exhibit noticeable decreases in reliability, suggesting the presence of potentially abnormal market conditions or data quality issues. Overall, the results indicate that the dataset is largely consistent and suitable for volatility forecasting.

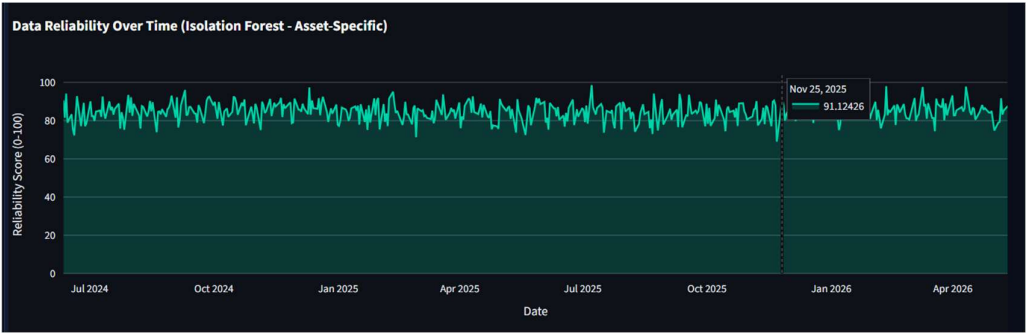


Figure 6.3: The reliability score over time computed by the isolation forest

6.4 Final Dashboard Output

Figure 6.4 shows the final dashboard output of the proposed decision-support system. The dashboard presents the volatility prediction together with the Data Reliability Score, allowing users to evaluate both the predicted market condition and the trustworthiness of the underlying input data. This output supports risk-oriented decision-making rather than direct trading decisions.

7. DISCUSSION

7.1 Interpretation of Results

The experimental results indicate that Logistic Regression achieved the best overall performance among the evaluated models. This finding suggests that the relationship between the selected technical indicators and future volatility is largely linearly separable. More complex models such as Random Forest and Gradient Boosting did not provide substantial performance improvements, indicating that additional model complexity may not always lead to better forecasting results for this dataset. The results are consistent with previous studies that emphasize the importance of carefully engineered features and proper validation strategies in financial forecasting. The findings suggest that interpretable machine learning models can remain competitive when relevant market indicators are used as input features.

7.2 Contribution of Data Reliability Assessment

A key contribution of this study is the integration of a Data Reliability Score into the forecasting pipeline. Unlike conventional volatility forecasting systems, the proposed framework evaluates the quality of incoming data before presenting predictions. This additional layer does not directly improve classification accuracy; however, it provides valuable context regarding the trustworthiness of the prediction and supports more informed decision-making. The reliability assessment component enables users to distinguish between predictions generated from highly reliable data and those produced from potentially abnormal observations. As a result, the proposed system provides not only a volatility forecast but also additional information regarding the quality of the underlying input data.

7.3 Limitations and Future Work

Several limitations should be acknowledged. First, the study focuses exclusively on the NASDAQ Composite Index and therefore the results may not generalize to other financial markets. Second, only historical OHLCV data and technical indicators were used, while macroeconomic variables, news sentiment, and alternative data sources were not considered. Future studies may investigate the impact of these additional information sources and evaluate the proposed reliability framework on different financial assets and forecasting horizons. In addition, advanced machine learning and deep learning models may be explored to determine whether further improvements in forecasting performance can be achieved.

8. CONCLUSION AND RECOMMENDATIONS FOR FUTURE WORK

8.1 Conclusions

The thesis demonstrates that day-ahead NASDAQ volatility can be successfully predicted to a statistically significant degree by using relationships between price and volume over time, technical indicators, and a score indicating data reliability. For the test period between 2020 and 2024, with $N = 756$ observations, Logistic Regression obtained the highest accuracy (70.63%, true positives over all instances) and achieved recall on true volatility peaks (70.99%), which could be useful in risk management, as failing to predict a volatility peak is more costly than incorrectly predicting one. The success of Logistic Regression indicates that the 22 technical indicators created a signal that was linearly separable to a large degree. An important secondary result is a reliability score based on an Isolation Forest, which can be used to filter out potentially erroneous data points. This provides additional information about data quality and helps users assess the trustworthiness of model outputs.

8.2 Theoretical and practical implications

The contribution of this paper can be situated between market efficiency and useful predictability; it demonstrates that statistical patterns of short-term price variation can be learned from price and volume series through careful feature engineering and validation. From a theoretical viewpoint, the findings are consistent with the phenomenon of volatility clustering and with the assumption that financial time series may contain exploitable structure, at least in the context of risk forecasting. From a practical viewpoint, two types of outputs are offered for a given prediction: (1) a volatility class prediction and (2) a Data Reliability Score ranging from 0 to 100. These outputs support tiered decision-making, allowing stronger reactions in high-confidence cases and more conservative action in low-confidence cases. In this sense, the system is better suited as a tool for monitoring risk than as an automated trading system.

8.3 Limitations

Even though the results were positive, the current model has various drawbacks: Firstly, the model was trained and tested solely on NASDAQ data between 2010 and 2024, and the generalizability to other indexes, asset types or geographical markets is unknown. Secondly, the model only uses price and volume information. Exogenous variables like macroeconomic data, implied volatility data or intraday data are not considered but may add explanatory power. Thirdly, only one look-back period is used and a split based on time sequence is applied to the training and test data. The effects of changing market regimes were not explicitly investigated.

8.4 Recommendations for further research and operational deployment

8.4.1 Adaptive Validation and Monitoring

In terms of the validation framework, it would be beneficial to transition from a static validation setup to a walk-forward system with a recurring retraining mechanism. This is due to the non-stationary nature of financial markets where models may perform poorly due to shifting market regimes. A monitoring system for both performance drift and data drift would enable a more robust system in practice.

8.4.2 Operationalizing Reliability Scores

The Isolation Forest-based reliability score could be implemented directly into the decision pipeline. The system can be set to act on the highly reliable predictions (e.g. Automatically), pass the moderate reliable predictions for human evaluation, and pass low reliable predictions to either be ignored or flagged for inspection. This would provide for a more practical system that fits the risk management processes.

8.4.3 Cross-Market and Multimodal Validation

This methodology could be implemented for other indices such as S&P 500 and Russell 2000. Other classes of assets like cryptocurrencies, commodities, and foreign exchange could also be used to implement the model in future studies. Macroeconomic and microstructure variables could also be added to future model iterations. This would ensure a better modeling of tail risk and the surrounding market environment.

8.4.4 Advanced Modeling and Hyperparameter Optimization

Future studies may evaluate advanced machine learning and deep learning models, together with systematic hyperparameter optimization techniques, to further improve forecasting performance.

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